

AI in Service Design: A New Framework for Hybrid Human–AI Service Encounters

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Abstract

The integration of Artificial Intelligence (AI) in service provision has prompted a revaluation of how we design service encounters, given the emergent role this technology plays in services. This article introduces the concept of the Hybrid Service Encounter to explore the evolving interplay between humans and AI in service contexts. We propose a 2×2 framework that categorizes service interactions into four distinct quadrants based on whether the provider and user are human or AI. Drawing on service design literature and current developments in AI, we analyze the implications of these hybrid encounters for service design roles, processes, and outputs. The article further identifies key areas for future research to guide the development of human-centered and ethical AI-powered services. Our contribution extends service design theory and practice by offering a framework to guide the integration of AI in service encounters.

Keywords

artificial intelligence, service design, service encounters, human–AI interactions

Introduction

The integration of Artificial Intelligence (AI) into service ecosystems is reshaping how services function and how they are designed. While AI has long supported back-end efficiency, nowadays it increasingly influences front-end interactions, raising fundamental questions about the future of service design. This article builds on the established body of service design research and extends it by introducing a conceptual framework—the Hybrid Human-AI Service Encounter—that enables researchers and practitioners to make sense of the new types of interaction AI introduces into service contexts.

Service design has matured significantly over the past 2 decades, characterized by an emphasis on co-creation, systems thinking, and the orchestration of multiple touchpoints across service journeys (Kimbell 2011; Meroni and Sangiorgi 2016; Wetter-Edman et al. 2014). This domain foregrounds user experience and reflexivity, typically drawing on methods like blueprinting, journey mapping, and prototyping (Bitner, Ostrom, and Morgan 2008). These elements remain highly relevant as services evolve to integrate new technologies. Yet, as AI begins to play an increasingly active role in both frontstage and backstage service environments, traditional assumptions embedded in these methods need to be re-examined. The current transformation is not simply about integrating AI into existing service systems; it is about AI altering how services are constructed, delivered, and experienced. For instance, in home maintenance, virtual assistants now go beyond adjusting lighting or temperature. Algorithms can monitor behavior, order groceries, or optimize appliances to reduce energy use. These

systems no longer merely support humans; they assume responsibilities and shape social behavior. As users begin to perceive algorithms as collaborators, service designers are challenged to consider a variety of new interactions, where both humans and AI can take on decision-making roles, fundamentally reshaping what designers must anticipate and configure.

Despite increasing attention to AI in service marketing and innovation (Bagozzi, Brady, and Huang 2022; Bock, Wolter, and Ferrell 2020; Haenlein and Kaplan 2019; Huang and Rust 2018; Li et al. 2021; Ostrom, Fotheringham, and Bitner 2019; Rafaeli et al. 2017; Robinson et al. 2020), less focus has been placed on how this technology reshapes foundational constructs of service design, particularly the roles of service actors, the nature of service interactions, and the processes of service delivery (Patrício, Gustafsson, and Fisk 2018). Services have long been treated as complex, multi-actor experiences (Patrício et al. 2011), yet the role of non-human agents is still under-explored (Rodrigues, Holmlid, and Blomkvist 2021). Furthermore, while service design has addressed social change (Sangiorgi 2011) and increasingly engages with complex systems and service ecosystems (van der Bijl-Brouwer 2022; Vink et al. 2021), it has yet to fully account for the implications of algorithmic agency, defined as the capacity of AI to act (Andrada, Clowes, and Smart 2023;

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van de Poel 2020). Like human service providers, AI systems have objectives and decision-making capabilities; unlike humans, they do not rely on social intuition but operate based on pre-trained models and engineered objectives (Huang and Rust 2022). This co-existence of human and AI agency thus necessitates closer attention to how decisions are shared, delegated, or negotiated, ultimately influencing user experiences and the orchestration of service elements. As a result, service designers need to develop new approaches to designing for hybrid (human–AI) service systems.

The first goal of this article is to review current knowledge on AI in services and to propose a new typology that captures how humans and AI interact in service encounters. Specifically, we introduce a 2×2 Human–AI model that considers both actors—humans and AI—as potential service users and providers. By interpolating these roles, the model defines four types of service encounter and identifies the implications that researchers and practitioners must consider to better account for these dynamics in service design. This article contributes to service design by offering a structured lens to understand human–AI interactions, discussing how AI reshapes core design concerns, and proposing a research agenda grounded in service design epistemology. In doing so, we support the development of design knowledge and practices capable of navigating the emerging opportunities and challenges of AI.

The remainder of the article unfolds as follows: we first define AI in relation to service design, introducing the concept of *Service AI* and outlining how this alters key elements of services across states and environments. We then present the four types of human–AI interaction identified in our framework and connect them to corresponding research questions. The article concludes with a discussion of the implications for the roles, processes, and outputs of service design, providing a foundation for future scholarship and practice within this evolving service landscape.

Defining AI for Service Design

In service design, services are shaped through the deliberate orchestration of systems, interactions, and experiences. The literature highlights how service designers conceptualize value through the integration of people, processes, and touchpoints across visible and invisible layers of delivery (Bitner, Ostrom, and Morgan 2008; Patrício et al. 2011). When AI enters this configuration, it affects not only how services are delivered but also how they are conceived and experienced. Consequently, defining AI for service design requires contextualizing its role within the layered complexity of services. Unlike traditional service technologies, AI does more than executing tasks; it interprets, learns from, and acts upon data, increasingly with autonomy. These capabilities challenge conventional boundaries between human and machine actors in services, requiring a renewed understanding of how AI influences both the environments and states within which services are designed.

Designing Services in a Dual Environment and AI

Services are designed across two environments: backstage and frontstage (Bitner 1992; Glynn Mangold and Babakus 1991). In the backstage, processes are established to facilitate service delivery. This involves integrating resources and strategically aligning three key factors: the firm's service concept as a value constellation of service offerings, the firm's service system comprising its architecture, collaborations, and workflows, and the service encounter shaped by the material evidence of the service (Roth and Menor 2003; Patrício et al. 2011). Each component influences the frontstage, where users interact with service providers to co-create the service experience (Glynn Mangold and Babakus 1991; Golder, Mitra, and Moorman 2012).

AI affects both domains simultaneously. For the organization providing the service, AI can make processes more efficient, for instance by automating repetitive tasks or managing delivery. Beyond being a tool executing predefined processes, AI can act as an autonomous system making data-driven decisions, optimizing workflows, and anticipating service failures. For end users, AI increasingly shapes experiences through recommendation systems, conversational agents, and intelligent-adaptive service interfaces that offer context-aware interactions. In both cases, AI contributes not only as a functional tool but also as an actor capable of autonomous decision-making, challenging the anthropocentric assumptions of service interaction (Bitner 1992; McCallum and Harrison 1985; Solomon et al. 1985). This shift repositions AI from a supportive technology to an agent, actively shaping the service encounter alongside human actors, thus demanding new service design approaches that accommodate multiple types of reasoning—human and computational—and acknowledge AI's potential to both mediate and lead interactions.

Designing Services in a Dual State of Being and AI

Service designers operate across two primary states of services: the potential one, where the service is conceived and planned, and the kinetic one, where the service is enacted (Shostack 1982). In the potential state, designers use models such as service blueprints and user journeys to map anticipated interactions, define roles, and sequence activities. In the kinetic state, the service is delivered and shaped by user actions, feedback, and contextual variables. One key method for managing both states is *service blueprinting* (Bitner, Ostrom, and Morgan 2008), which provides a structured approach to visualizing service processes, user interactions, and corresponding organizational activities. By distinguishing between visible (frontstage) and invisible (backstage) components, blueprints help illuminate the service journey and its underlying mechanisms. However, traditional blueprints are based on the assumption that service execution follows scripted rules and predefined sequences of activities. This logic is disrupted by AI, which introduces adaptive, data-driven, and self-learning architectures that

evolve in response to user behavior and environmental input (Xu et al. 2020). In AI-powered services, humans and machines co-shape the service experience. Thus, designers are challenged with accounting not only for predefined scenarios but also for emergent interactions—particularly those involving AI systems acting autonomously. Static blueprints fall short in capturing these complexities, especially when interactions span human-to-human, human-to-AI, and AI-to-AI exchanges.

To illustrate, consider AI-powered mobility services such as ride-hailing platforms or autonomous shuttle systems. In the potential state, designers might plan user journeys, define the logic for dispatching vehicles, and script the interaction points through which users request rides and receive updates. However, once deployed, the kinetic state is shaped by AI systems that make continuous, autonomous decisions in real time—reassigning vehicles, rerouting based on traffic patterns, and dynamically adjusting estimated times of arrival based on incoming data. These systems are not simply executing pre-scripted tasks; they are co-producing the service through ongoing data interpretation and autonomous decision-making. Such capabilities challenge the assumptions behind static blueprints and require new design tools capable of accommodating uncertainty, emergent behaviors, and machine agency.

A Definition of Service AI

The emergence of AI in service contexts calls for a definition that aligns with the principles, tools, and values of service design. The definitions of AI currently offered in the service literature fail to capture the depth of AI's role described in the previous sections (a summary of definitions is provided in Table 1). Defining AI in contrast to human intelligence (Huang and Rust 2018; Ostrom, Fotheringham, and Bitner 2019; Rafaeli et al. 2017; Syam and Sharma 2018) or in terms of functional capabilities (Bagozzi, Brady, and Huang 2022; Bock, Wolter, and Ferrell 2020; Haenlein and Kaplan 2019; Li et al. 2021; Robinson et al. 2020) tells us little about the implications for service design, and the extent to which AI can co-create experiences. According to service design epistemology, emphasis should be given to AI's contribution to co-creating interactions, experiences, and systemic coherence across the frontstage and backstage (Kimbell 2011; Patrício et al. 2011; Shostack 1977). The dynamic configurations of people, processes, technologies, and touchpoints should also be emphasized, acknowledging that in AI-powered services, human-to-AI and AI-to-AI interactions play a role as central as human-to-human interactions. Against this backdrop, AI should be understood not merely as a technology that enables automation or personalization, but as an agent that co-creates value (van de Poel 2020) across different service environments and states. To account for this agency, two defining characteristics need to be introduced: (1) *hybridity*, highlighting that service interactions happen in hybrid configurations, where AI and humans can each act as users or providers of a

service with AI possibly replacing people; (2) *co-piloting*, underscoring the agency of AI in co-shaping service journeys with humans and other AI agents.

We thus propose a definition of AI for service design (or Service AI) as technology that co-creates service value and co-pilot service delivery with both users and other machines across frontstage and backstage environments. This definition acknowledges AI's proactive role in service encounters: by conceptualizing it as a co-pilot rather than a passive tool, it positions AI as a contributor to value creation. It also foregrounds the hybrid nature of modern service ecosystems, where humans and AI can both act as users and providers, challenging traditional linear models. Our definition also aligns with recent service design discourse on systemic complexity and service ecosystems (van der Bijl-Brouwer 2022; Vink et al. 2021) and provides a foundation for future research and practice that addresses how AI reshapes the principles, methods, and outcomes of service design within multi-agents environments.

Service AI Shaping a Hybrid Service Encounter

Based on our definition of Service AI, we posit that there are multiple ways in which humans and AI can co-create and co-pilot services. While theories have traditionally depicted services as exchanges between humans (McCallum and Harrison 1985; Solomon et al. 1985), Service AI introduces a conception where value co-creation can encompass a blend of human-to-human, human-to-AI, and AI-to-AI interactions, both in the front- and backstage. Furthermore, akin to humans who can function as both service providers and recipients, AI assumes a dual role: algorithms can autonomously deliver services, as many service operations are increasingly handled exclusively between algorithms, or function as recipients of services. With Service AI, the concept of hybridity becomes central also in service design. An illustration of this can be made using the example of a user-facing chatbot: human-to-AI interactions occur when users engage with a chatbot to resolve their inquiries. The chatbot provides responses either through predefined scripts or interacting in human-like ways via natural language processing capabilities. However, human-to-human interactions may also come into play if the chatbot fails to adequately address the user's query. In such cases, the interaction shifts from human-to-AI to human-to-human, with a person taking over the conversation. Behind the scenes, AI-to-AI interactions can also occur where the chatbot communicates with other AI systems to access information or cross-check data.

These hybrid configurations have direct implications for service design. Traditional tools such as service blueprints and journey maps must be adapted to account for non-human actors with decision-making capabilities (Patrício et al. 2011). For instance, mapping hybrid interactions requires designers to consider the logic and constraints of AI systems, such as data flows and algorithmic decisions' impact on user experiences. Moreover, hybrid configurations signify a shift in the conception of service

Table 1. Overview of the Definitions of AI in Service Research.

Author(s)	AI Definition	Application in the Service Sector
Rafaeli et al. (2017)	Technologies that include hardware, software, and information	Three-generational technology innovation including: (1) use for standardization, (2) replacement or augmentation of frontline employees, and (3) facilitation
Huang and Rust (2018)	Machines exhibiting facets of human intelligence	Four types of AI in services: mechanical, analytical, intuitive, and empathetic
Syam and Sharma (2018)	The ability of machines to mimic intelligent human behavior, and specifically “cognitive” functions that we associate with the human mind, including problem-solving and learning	Various applications in sales, from predictive modeling to information extraction and summarization
Ostrom, Fotheringham, and Bitner (2019)	Nonbiological intelligence	Three types of applications: AI-supported, AI-augmented, and AI-performed services
Haenlein and Kaplan (2019)	A system’s ability to interpret external data correctly, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation	Three main applications of AI: (1) the relationship between firms and employees, (2) the changes to the internal functioning of firms, (3) the interaction between a firm and its customers
Bock, Wolter, and Ferrell (2020)	The configuration of technology to provide value in the internal and external service environments through flexible adaptation enabled by sensing, learning, decision-making, and actions	AI influences service quality, relationship marketing, social exchange, service failure, service-dominant logic (SDL), emotional contagion, servicescape, service profit chain, and service climate
Xu et al. (2020)	A technology-enabled system for evaluating real-time service scenarios using data collected from digital and/or physical sources in order to provide personalized recommendations, alternatives, and solutions to customers’ enquiries or problems, even very complex ones	AI in customer service
Robinson et al. (2020)	AI has the ability to learn, process, affect, and mimic human characteristics	Service encounters involving AI can be of three types: human-to-human, human-to-AI, AI-to-AI
Li et al. (2021)	AI technologies differ from other nonhuman factors in terms of anthropomorphism and intelligence to such a degree that they can be regarded as automated social presence	Four modes of AI technology-based service encounters: AI-supplemented, AI-generated, AI-mediated, and AI-facilitated encounters
Bagozzi, Brady, and Huang (2022)	A feeling machine for serving customers’ emotional needs	Interacting with AI can generate three categories of emotion: basic, self-conscious, and moral emotions
Mortati and Viana Mundstock Freitas	Service AI is technology that co-creates service value and co-pilot service delivery with both users and other machines across internal and external environments	Embedding AI in services generates four types of service encounter: human-to-human, human-to-AI, AI-to-human, AI-to-AI

encounter that our framework conceptualizes while also prompting future research through the identification of a range of pertinent questions.

Exploring the Hybrid Human–AI Service Encounter

Our framework (Figure 1) defines four types of interactions between humans and AI in services, acting as a tool for analyzing

how humans and AI co-create and co-pilot services, while highlighting the distinction between front- and backstage activities (outlined in Table 2). In the following sections, we provide a description for each quadrant of the framework illustrating examples and discussing the novelties that Service AI brings. Exemplar topics and research questions are also identified and examined for each quadrant (Table 3). Finally, implications for service design are discussed according to roles, processes, and outputs (Table 4). In doing this, our aim is not to provide an exhaustive

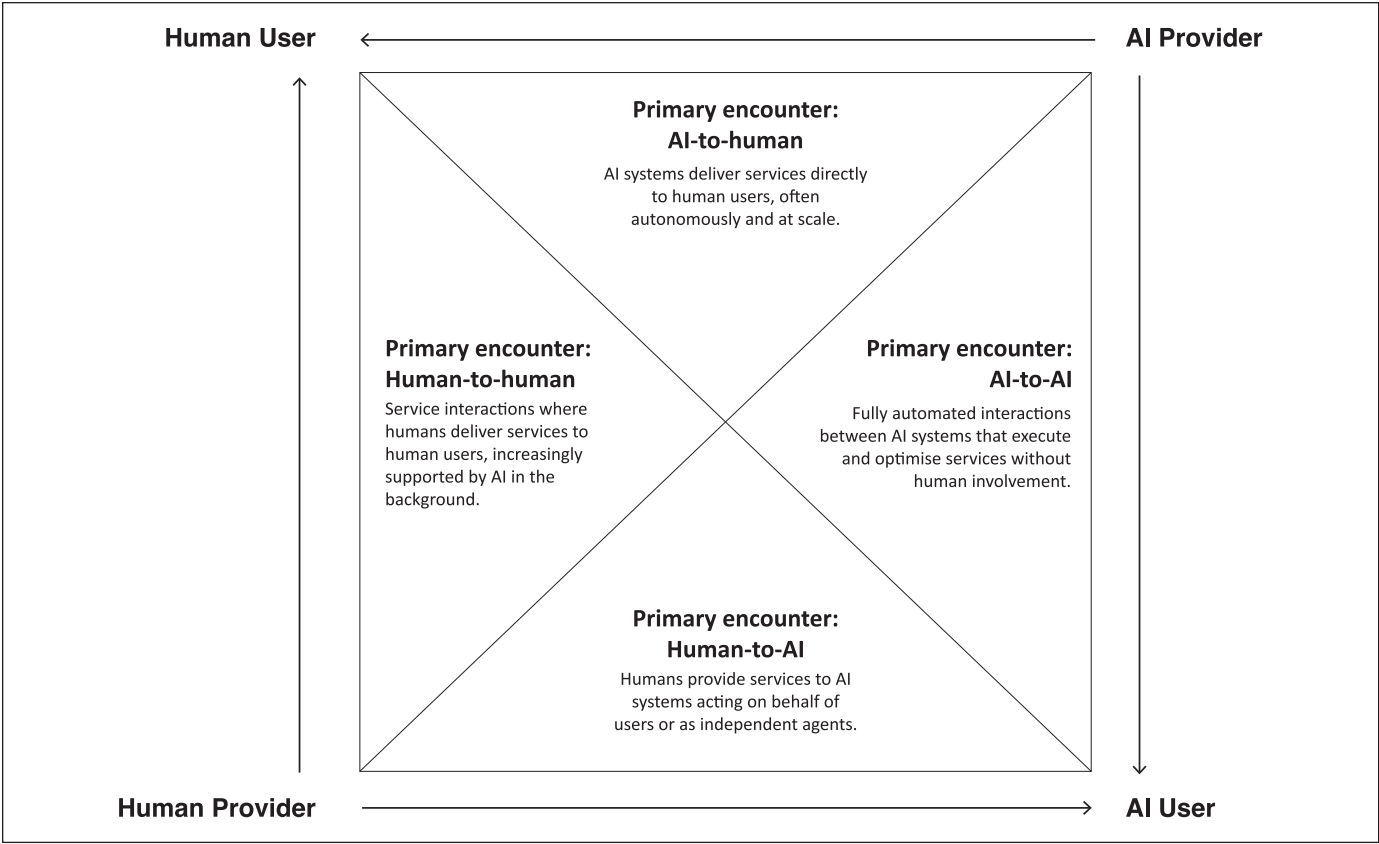


Figure 1. Four types of interaction in the hybrid Human–AI Service Encounter.

Table 2. Characteristics of Frontstage and Backstage in Relation to Service AI.

Service environment	Primary Type of Exchange in the Hybrid Service Encounter			
	Human-to-Human	Human-to-AI	AI-to-Human	AI-to-AI
Frontstage	Humans primarily interact with other humans. Effective for services requiring intimate social interactions	AI interacts on behalf of humans making decisions on service use	Humans interact with algorithms via the service interface, co-piloting service delivery	AI receives inputs from other algorithms in services devoid of interpersonal interactions
Backstage	Humans and AI collaborate to co-deliver the service. AI elaborates data to support human activities	Humans operate making critical choices on service processes, making them understandable to AI	AI operates on behalf of (or in cooperation with) humans and dynamically adjusts the service journey based on continuous data analysis	AI receives and elaborates data to feed the activities of other machines

list of research questions and implications; rather, we offer a starting point for advancing knowledge in the field.

Human Provider to Human User

The human-to-human quadrant depicts a primary encounter where both the service user and provider are humans. This represents a common situation in service design that has been

extensively studied in prevailing theories and that will continue to be relevant in many services, particularly those that heavily rely on social interactions. According to scholars, human-to-human service exchanges are characterized by high levels of affect and a need for intimate relationships (Robinson et al. 2020). These factors highlight why this type of interaction is often challenging for integrating AI into frontline roles, despite recent frameworks researching the

Table 3. Research Questions Across the Four Types of Human-AI Service Encounters.

Encounter Type	Key Areas of Investigation	Exemplar Research Questions
Human-to-human	Service AI is a hidden partner in service delivery	(1) <i>User experience and interaction protocols</i> How does hidden AI reshape human-to-human service delivery, and what degree of visibility is appropriate? (2) <i>Ethical and responsible design practices</i> How can bias and unintended consequences be mitigated when AI works in the background? (3) <i>Transparency and algorithmic fairness</i> How can designers make the presence and role of AI intelligible and trust-enhancing for users? (4) <i>Agency and empowerment</i> How can service designers balance the need for AI-driven personalization with concerns about user privacy and data protection? How can human agency be preserved when AI systems influence or mediate delivery?
Human-to-AI	Service AI is the main customer of services	(1) <i>Optimization of interaction protocols</i> What forms of service interaction do AI systems require when acting as users? (2) <i>Interfaces designed for AI systems</i> How should interfaces be optimized for algorithms to make informed decisions? (3) <i>Algorithmic fairness and bias mitigation</i> How can service designers mitigate algorithmic biases and contribute to ensuring fairness in AI decision-making processes?
AI-to-human	Service AI is used to tailor experiences at scale	(1) <i>Ethical implications of personalization</i> Are there parameters or guidance that can be established for service designers to build the scaffolding for Service AI? How can service designers create ethical data feedback loops to support algorithmic functions? (2) <i>Long-term effects of personalization on user behavior</i> What are the long-term effects of AI-driven personalization on user behavior, such as engagement and trust in service providers? (3) <i>Optimizing personalization algorithms</i> How can service designers optimize algorithms to deliver more accurate, relevant, and context-aware personalized experiences?
AI-to-human	Service AI is used to enhance user engagement	(4) <i>AI-driven engagement strategies</i> What are the key factors that contribute to the success or failure of AI-driven engagement strategies, and how do they vary across different user demographics and service domains? (5) <i>Balance between human–AI control</i> What are ways in which a good balance can be ensured between human control over choice and AI control over recommendations? (6) <i>Measuring and evaluating user engagement</i> How can service designers effectively evaluate the impact of AI-driven engagement strategies?
AI-to-AI	Service AI acts on behalf of humans	(1) <i>Managing transparency and explainability</i> What mechanisms can be implemented to provide users with agency over how AI makes decisions? (2) <i>Designing for user empowerment and control</i> How can service designers empower users to retain control over AI-driven service interactions? (3) <i>Ensuring accountability and responsibility</i> To what extent is it ethical that humans remain blind to AI exchanges? How can service designers design AI-driven services to facilitate traceability and auditability of AI actions?
	Service AI infers human behavior	(4) <i>Exploring the impact of inferred behavior</i> How does the incorporation of AI-inferred human behavior impact the design of services? (5) <i>Focusing design activities on evaluation</i> How do the role and skills of service designers change when their task is linked to guiding and monitoring a machine? (6) <i>Long-term and socio-cultural impacts</i> What are the long-term implications of widespread adoption of AI-inferred behavior on societal norms, cultural practices, and autonomy? How can service designers anticipate and mitigate unintended consequences or societal biases resulting from the use of AI-inferred human behavior in service design?

Table 4. Implications Shaping the Future of Service Design in Relation to Service AI.

Quadrant	Service Designer's Role	Service Design Process	Service Design Outputs
Human-to-human (AI in the backstage)	Designers must assess where and how AI remains hidden, balancing efficiency with transparency. Their role involves determining the visibility of AI and shaping user expectations	Designers integrate AI to automate tasks, or augment decision-making without altering the traditional human-facing service journey. They must be aware of how AI silently influences outcomes	Outputs remain primarily human-driven but may be AI-supported in the backstage. Designers must critically evaluate how AI alters services even when users do not directly interact with it
Human-to-AI (AI as service user)	Designers act as interpreters and facilitators of AI needs, determining which data, rules, and goals are necessary for AI. Their role becomes one of translating human logic into machine-readable formats	The process emphasizes data architecture, interaction protocols, and context modeling. Designers work upstream to train, instruct, and contextualize AI, treating it as a new kind of service actor	Outputs focus on optimizing AI understanding and responses, including structured datasets, rules for inference, and interfaces readable by machines
AI-to-human (AI as service provider)	Designers co-pilot service delivery with AI, guiding how the system generates, communicates, and adapts outputs. They assume roles of editors, curators, and orchestrators of AI-generated experiences	Design methods evolve to include prompting, prototyping with AI, and evaluating AI-generated alternatives. Iteration cycles involve both human judgment and AI feedback	Outputs are co-created by AI and designers, ranging from personalized content to adaptive interfaces. Designers are responsible for refining, contextualizing, and aligning these with user needs and values
AI-to-AI (AI in both user and provider roles)	Designers oversee how AI systems interact and what kinds of decisions are delegated. Their role includes establishing protocols for inter-AI communication and auditing interactions	Designing implies orchestrating systems, data flows, and communication rules between AI agents. Designers ensure technical systems are interoperable and ethical	Outputs include service architectures, governance protocols, and interaction rules between AI systems. Although often invisible to humans, these are critical to the service experience

integration of AI for empathetic tasks (Bagozzi, Brady, and Huang 2022). Here, the changes that Service AI introduces concern primarily the extent to which service providers choose to utilize AI as a support tool in the backstage, and the nature of the hybrid collaboration that may emerge. To illustrate, in a telemedicine application, AI can aid doctors during consultations with patients. When a patient interacts with the telemedicine platform, the AI system can assist by transcribing spoken patient information, analyzing medical records and prior diagnoses, and suggesting treatment options based on symptoms and medical history. Throughout the virtual consultation, the AI system can enable quick access to patient data, perform real-time analysis, and offer evidence-based recommendations. This can significantly enhance service quality, ensuring more precise diagnoses, efficient treatment plans, and an improved overall patient experience. In such applications, Service AI has the potential to supplant individuals in various roles, streamlining service delivery while the primary interaction remains between doctors and patients (Li et al. 2021; Xu et al. 2020).

These applications raise several novelties for service design, such as the impact that the human–AI collaboration in the back-end might have on the service outcome and the need to disclose

AI presence to end-users. The trust and acceptance that individuals have towards the presence of AI in the backstage is an increasing concern for designers, who need to balance the distribution of agency and effectively communicate the tasks of the technology.

Human Provider to AI User

The human-to-AI quadrant refers to situations where AI systems, either partially or fully autonomously, interact with service providers on behalf of human users or in place of them altogether (Huang and Rust 2022). In these cases, AI systems perform tasks typically carried out by humans, such as making purchase decisions or initiating service requests, thus becoming the main users in the frontstage of services. This transformation is visible in both business-to-business and business-to-customer service types. For instance, in business-to-business settings, AI systems may autonomously manage inventory optimizing stock levels and ensuring timely deliveries. In business-to-customer settings, AI technologies like virtual assistants can schedule appointments, make purchases, or engage with service interfaces without human intervention. In both cases, AI leverages its ability to forecast demand and supply, and relieves humans

from tedious tasks, like booking appointments or grocery shopping. The critical differentiator of this typology is when AI makes service use and purchase decisions on its own, thus becoming a service user.

This encounter, although relatively less studied, is not new in service literature. Robinson et al. (2020) describe a notable example with Google Duplex, where AI can interact with human providers in a manner akin to a real person. Meanwhile, Huang and Rust (2022) identify three levels of AI customer: (1) AI augments human customer, (2) AI replaces human customer, and (3) AI is the customer. An example can be illustrated through a hotel reservation system where a virtual assistant autonomously books a hotel room on behalf of a person. The AI interacts with the hotel's booking agent as if it were a human, negotiating room availability, confirming reservation details, and finalizing payment. From the perspective of the hotel staff, the AI becomes the user, although mimicking a human. In the background, the AI system performs a range of tasks, including analyzing the preferences of the human user, comparing prices across multiple platforms, and selecting the optimal booking option.

This interaction highlights several key implications for service design, including the need to rethink interaction protocols to accommodate AI systems that do not require emotional responses or conversational nuance. Instead, service interfaces need to be optimized for human-to-machine communication, ensuring that AI systems are granted machine-readable access to relevant data. Furthermore, designers need to navigate the implications of AI autonomy, determining which permissions and boundaries should govern AI actions, and ensuring that service environments are prepared to verify and respond to algorithmic decisions.

AI Provider to Human User

While interactions wherein humans provide services to AI users are relatively novel, services delivered by AI to human users are now widespread. These include a range of offerings where the provider is primarily an algorithm (or an ensemble of algorithms) capable of responding to human requests. Services like Amazon and Netflix exemplify this, with algorithms operating across front- and backstage.

This interaction has been extensively studied (Glushko and Nomorosa 2013; Hollebeek and Belk 2021; Huang and Rust 2018; Li et al. 2021; Ostrom, Fotheringham, and Bitner 2019), with research exploring how service design methods are being adapted to address AI-led service delivery (Yang, Banovic, and Zimmerman 2018; Yildirim et al. 2022). A key development is the collaboration between humans and algorithms, which expands interface diversity, capacity, and functionality (Van Doorn et al. 2017). This is evident in natural language-based interfaces that enable human-like exchanges and dynamically adjust to user needs. In some cases, Service AI entirely replaces human providers. For instance, fitness apps can

now generate adaptive workout plans by analyzing users' fitness levels, goals, and preferences. These apps collect ongoing performance data and refine their suggestions in real time, delivering personalized services without human involvement. Such examples illustrate how AI modifies both backstage processes (data analysis and decision-making) and frontstage experiences (user interactions), enabling personalization at scale (Glushko and Nomorosa 2013).

These innovations raise critical implications for service design, including addressing the challenges of algorithmic differentiation between users (e.g., distinguishing an adult from a child) and the delegation of service decisions to non-human agents. Service AI follows different cognitive logics, requiring design principles that account for its limited understanding of emotional nuance and context (Lee, Hosanagar, and Nair 2018). As Girardin and Lathia (2017) note, addressing these issues is vital to ensure Service AI supports, rather than undermines, human well-being.

AI Provider to AI User

In this type of encounter, algorithms interact autonomously without human intervention, involving primarily technology and information systems. This aligns with what Glushko (2010) termed information-intensive services, where value is generated through the exchange of data (e.g., documents, text, videos) rather than through interpersonal interactions. Unlike services involving human contact, these automated exchanges are devoid of social components and have become commonplace with the rise of smart service systems featuring self-detection and self-diagnostic technologies (Maglio 2014). While in the past these activities were confined to the back-office, they are now increasingly moving to the frontline of services, with people as the ultimate beneficiaries and the activities primarily unfold between machines. For instance, consumer appliances equipped with sensors and connectivity can anticipate user needs and autonomously place orders to manage routine tasks. This means that while in the backstage data is shared, in the frontstage AI systems make decisions on service implementation. Robinson et al. (2020) call this type "interAI" services, where AI agents communicate with one another. One example can be an automated system for traffic monitoring and regulation, where algorithms continuously analyze traffic patterns, predict congestions, and adjust traffic lights or speed limits to keep vehicles moving smoothly, all without human intervention. Another is self-driving vehicles that communicate exclusively with other algorithms to perform their tasks.

The implications of this encounter for service design are manyfold. Frameworks that emphasize human interactions need to be rethought to integrate the facilitation of AI-to-AI communication across service environments. In the backstage, this involves creating systems for data sharing and integration to enable algorithms' operations. In the frontstage, this implies accounting for algorithmic autonomous decision-making while safeguarding human oversight through appropriate interfaces and infrastructures that align AI operations with ethical standards.

Advancing Service Design: Research Directions on the Hybrid Service Encounter

The Hybrid Service Encounter framework offers a structured lens for exploring how the integration of AI is reshaping the foundations of service design. To examine these shifts more specifically, this section presents a research agenda grounded in the epistemology of service design research and practice—particularly its emphasis on co-creation, systems thinking, and reflexivity. These values have long distinguished service design from adjacent disciplines and provide a critical foundation for examining the implications of AI. Co-creation plays a central role in service design, reflecting its ethos of involving users and stakeholders as active contributors in the service development process (Björgvinsson, Ehn, and Hillgren 2012; Manzini and Rizzo 2011). The introduction of AI, however, challenges this principle by introducing a non-human co-designer and shifting agency. When AI becomes an agent in the co-creation of services, the question of who the users are needs to evolve accordingly, prompting designers to rethink how participatory methods can be adapted to include algorithmic participants and support hybrid forms of collaboration. Systemic thinking encourages designers to view services as part of complex socio-technical systems (Bauer and Herder 2009; Li et al. 2020). This lens becomes especially relevant in the context of Service AI, which operates across platforms, data infrastructures, and organizational boundaries. Designers must now consider how to represent and intervene in service systems where AI holds distributed agency, interacts with other technical components, and shapes user experiences in ways that are not always visible or predictable. Reflexivity concerns the designer's critical awareness of their own position, intentions, and influence on their solutions (Pihkala and Karasti 2016; Vink and Koskela-Huotari 2022). As designers work alongside AI, this awareness becomes even more necessary to shape responsibly how AI systems behave, make decisions, and mediate human interactions. Meanwhile, important ethical and methodological questions around responsibility, transparency, and value alignment become crucial. Together, these commitments highlight that Service AI is not simply a new tool but a transformative force that redefines the very foundations of service design. The four types of hybrid service encounters offer a conceptual scaffold for navigating these shifts and developing a research agenda that can help the discipline respond to the emergence of AI-powered services. The following sections present quadrant-specific research directions (see Table 3), each aimed at advancing knowledge and development of responsible design practices. However, it is relevant to stress that while the framework presents each quadrant as distinct, in practice these types often overlap and intersect within the same service journey—posing additional design challenges that must be addressed holistically.

Human Provider to Human User: Research Questions

In service encounters where both the provider and the user are human, AI often remains unseen, leading to critical design challenges. One of these concerns how hidden AI systems shape user experience. These systems may influence decisions and adjust the flow of interaction in ways that users neither request nor perceive. Designers, therefore, need new approaches to evaluate how AI modifies user perception and ensure that service outcomes remain user-centered, even when algorithms are driving key functions behind the scenes.

A second set of concerns revolves around ethical and responsible design. When AI makes or influences decisions that appear fully human-driven, the absence of visibility can complicate questions of fairness, inclusion, and accountability. Designers must develop strategies to critically assess whether the AI is reinforcing or mitigating bias and how to uphold ethical principles in opaque systems. Closely related to this is the challenge of transparency. Determining whether, when, and how to communicate the presence and role of AI is becoming a crucial part of service design. Transparency is not simply about revealing that AI is present, but also about helping users understand how their data is used and how decisions are shaped. Designers must strike a balance between clarity and overload, developing communication strategies that are accessible, contextual, and aligned with user expectations. Finally, invisible AI may shift the balance of agency: by automating decisions or streamlining processes without user input, AI systems can reduce users' sense of control. Designers must grapple with how to maintain human agency, ensuring that services remain empowering even when decision-making is partially delegated to machines.

Together, these concerns illustrate the importance of expanding service design practices to reveal and account for the hidden layers of AI. Research that addresses how to design with, around, and through invisible AI will be central to advancing human-centered services.

Human Provider to AI User: Research Questions

As AI systems increasingly act as service users, the new conceptual and methodological challenges that service designers are confronted with include redefining assumptions about how services are consumed, and how interactions are shaped.

One key area of concern is the optimization of interaction protocols between human service providers and AI systems. Traditional service interactions are designed around the needs, expectations, and abilities of human users; however, AI systems require structured, machine-readable inputs and respond to these based on predefined logic and learned behaviors. Accordingly, designers need to explore interaction protocols that allow AI systems to effectively navigate service environments. This may include the adoption of rule-based communication structures or

standardized formats for real-time information exchange. The aim is not only functional interoperability but also ensuring that AI systems can engage meaningfully with human providers without misinterpretation.

Closely related to this is the design of interfaces intended for AI use. Whereas most service interfaces are optimized for human cognition, interactions with AI systems may need to prioritize clarity, precision, and efficiency of data flow. Designers must investigate how service interfaces can be adapted to support algorithmic interpretation. This includes structuring service offerings, labeling content, and providing contextual metadata that AI systems can parse accurately. The designer's role thus becomes one of translation and scaffolding to make services legible and actionable for non-human users.

A third area of research is algorithmic fairness and bias mitigation. While designers have long grappled with fairness in human-facing services, designing for AI users introduces new complications. For instance, if an AI virtual assistant chooses a service provider based on biased data, human users may unknowingly be subject to discriminatory practices. Designers must develop new tools and frameworks to audit how service access is mediated through AI, ensuring that systems operate ethically and inclusively even when acting autonomously.

Finally, this quadrant raises questions about representation and accountability. When AI acts as a proxy for a human, whose preferences are it really enacting? Designers should thus consider how to ensure that AI users reflect the needs and values of the person they represent through allowing humans to shape and monitor how AI systems interact with services on their behalf.

AI Provider to Human User: Research Questions

The quadrant describing interactions where AI acts as the service provider and humans as the end-users is the most established (Hollebeek and Belk 2021; Huang and Rust 2018; Yildirim et al. 2022). AI systems are already widely deployed in this role, generating extensive implications for service design.

In this context, new research is needed to guide the development of systems that enhance, rather than compromise, the service experience. Focusing, for instance, on the ethical implications of personalization, AI systems can personalize service delivery based on vast amounts of personal data, but this often occurs without the user's full awareness or consent. This raises the need for service designers to explore how personalization can be implemented ethically, ensuring that tailored experiences do not undermine privacy.

Linked to this, research concerns also raise on the long-term effects of personalization. While AI systems may increase short-term engagement, they may also reinforce filter bubbles or foster dependency on algorithmic choices. Service designers must be aware of the broader consequences of prolonged interactions with AI and how these might reshape social behaviors.

Another area of investigation relates to the optimization of personalization algorithms themselves. Designers increasingly

collaborate with technical teams to define the inputs, thresholds, and feedback loops that shape the algorithm's decision-making. This involves determining which data points should be prioritized, how user feedback is integrated, and how adaptability is managed over time. Further research is thus needed to clarify the designer's role in shaping these systems and to develop methods for co-designing AI decision processes.

A fourth area is the development of AI-driven engagement strategies. These involve designing interfaces, touchpoints, and service journeys that maintain user interest and reduce friction. However, these seamless experiences may lead to unwanted consequences like digital addictions (Forlano and Mathew 2017), requiring designers to develop strategies that, while orchestrating meaningful experiences, preserve human values and integrity. Building on this, also balancing human and AI control is crucial. Here, research is needed to identify how to equilibrate automation and user empowerment, including exploring how much visibility and override capability humans should have.

Finally, an increasingly important area for investigation is the development of methods for measuring and evaluating user engagement in AI-powered services. Traditional service evaluation metrics may no longer apply when interactions are partially hidden. Designers need new frameworks to assess not only usability and satisfaction but also transparency, trust, and ethical alignment.

AI Provider to AI User: Research Questions

The final quadrant captures service encounters in which algorithms interact autonomously. While typically invisible to end users, these AI-to-AI interactions are crucial for enabling real-time responsiveness in service delivery. As such, they represent an important but underexplored frontier for service design.

In this quadrant, the first area for future investigation is the optimization of interaction protocols between AI systems. Designers must explore how to support autonomous exchanges between algorithms that operate with varying logics, data structures, and organizational goals. This includes participating in the standardization of communication mechanisms (e.g., APIs, metadata structures, ontologies) that allow AI systems to understand each other and coordinate actions seamlessly.

Second, the design of interfaces for AI systems represents a novel challenge. Whereas traditional interfaces mediate human-computer interaction, this quadrant requires designing interfaces that are machine-readable. These might consist of structured datasets, sensor-triggered protocols, or system-level integrations—requiring designers to think beyond touchpoints and instead collaborating in developing backend infrastructures.

A third area is algorithmic fairness. When AI systems interact and make decisions, there is a heightened risk of reinforcing and amplifying biases embedded in their respective training data. Designers must consider how to ensure that AI-to-AI

exchanges do not result in discriminatory or unethical outcomes, and how monitoring and auditing tools can be integrated into these invisible layers of service interaction.

The fourth area involves, once again, the balance between human and AI control. Even in scenarios where AI systems operate autonomously, human oversight remains essential. Designers must grapple with how to embed human-in-the-loop mechanisms at the system level, such as fail-safes, interpretability tools, or escalation pathways. Understanding how and when human intervention is appropriate in AI-to-AI systems is vital for maintaining accountability.

The fifth area concerns methods for measuring and evaluating performance. Given the lack of direct human experience in these interactions, traditional service evaluation metrics may no longer apply. New methods are needed to assess the quality, reliability, and ethical performance of AI systems interacting with one another, especially as their outcomes increasingly shape the services humans receive.

Finally, ethical and legal questions around data ownership and responsibility require further research. When AI-generated content or decisions are used in downstream applications, it remains unclear who owns the data, who is accountable for outcomes, and how such interactions can be traced and audited. Designers must anticipate these questions when developing autonomous systems, ensuring transparency, and ethical safeguards.

Implications of the Hybrid Service Encounter for Service Design

As outlined throughout the article, the Hybrid Service Encounter framework challenges conventional service design assumptions and practices. The research questions proposed underscore numerous implications needing further analysis. To discuss them, we organize this section around three central concerns: the role of service designers, the service design process, and the outputs of service design. These areas were selected to reflect research in service design. Specifically, the role of service designers aligns with literature exploring the evolving responsibilities and skills required as AI becomes embedded in services (Cautela et al. 2019; McCormack et al. 2020; Seidel et al. 2018; Wilson and Daugherty 2018). The service design process is informed by research on how AI adoption modifies design methodologies (Chromik, Lachner, and Butz 2020; Jylkäs et al. 2018; Kunneman, Alves da Motta-Filho, and van der Waa 2022; Main and Grierson 2020). Outputs of service design encompass explorations of the tangible and intangible results needed to depict AI-powered services (Huang and Rust 2018; Koch 2017; Yang, Banovic, and Zimmerman 2018; Yildirim et al. 2022). Within each of these domains, we examine how the hybrid encounter manifests across the four quadrants of our framework, articulating the unique implications of AI and identifying areas where traditional design practices must be rethought or adapted (summarized in Table 4).

Service Designers' Role

When shaping hybrid service encounters, designers must navigate a landscape where both humans and AI agents participate in service co-creation. As a result, their role evolves from that of creators to orchestrators and evaluators who collaborate with AI to co-pilot service creation and delivery (Feldman 2017). This shift has been interpreted in various ways across the literature: Koch (2017) describes designers as mentors imparting knowledge to AI; Wilson and Daugherty (2018) see them as analytical decision-makers; McCormack et al. (2020) position designers as expert judges evaluating algorithmic outputs; and Seidel et al. (2018) envision them as coaches guiding AI toward user-aligned results. This evolution also varies depending on the nature of the service encounter.

Human-to-Human. In this quadrant, where AI is embedded in the backstage, designers must assess to what extent AI systems should be visible or remain hidden. Their role involves determining how much of the AI's activity should be disclosed and how to communicate AI contributions to build trust and maintain transparency. Designers also need to manage users' perceptions and expectations, especially when decisions are influenced by invisible algorithms.

Human-to-AI. Here, designers must adapt to serving AI as a new kind of user. Their responsibilities shift toward designing interaction protocols and operational interfaces that allow algorithms to access and process information effectively. Designers must ensure that services are not only understandable to humans but also interpretable and usable by AI systems. This means creating service environments where data is structured in a way that machines can process, and where clear rules and contextual information are built in to guide the AI's decisions and actions.

AI-to-Human. In this exchange, designers focus on how humans interact with AI-generated outputs. The emphasis lies in ensuring that these outputs are meaningful, contextually appropriate, and ethically sound. Designers must become curators and interpreters, ensuring that algorithmic decisions make sense to users and that users are supported in navigating the AI's recommendations. This demands attention to fairness, explainability, and usability—especially in services where personalization and decision-making are delegated to algorithms.

AI-to-AI. In fully automated interactions, designers must support communication between algorithms. This includes contributing to designing data exchange protocols, managing interoperability between systems, and ensuring that decisions made by one AI are appropriately interpreted and acted upon by another. While these interactions may occur without human awareness, designers still play a critical role in shaping the rules, boundaries, and ethical safeguards that govern machine-to-machine exchanges.

Across all four encounters, the transformation in role also includes a broader shift from problem-context navigation to solution-context scaffolding (Cautela et al. 2019), and a demand for new capabilities such as prompt engineering, critical data analysis, and decision logic evaluation (Davenport and Mittal 2022; Yildirim et al. 2022). Designers are increasingly expected to collaborate with AI not only in generating but also in judging and refining algorithmic outputs—often contributing more to the orchestration of the design process than to its execution. Altogether these shifts suggest a multi-faceted transformation in service designers' responsibilities, which requires them to develop literacy in algorithmic operations, understand how to balance human and machine agency, and cultivate critical, ethical, and evaluative thinking. While this shift is still unfolding, further empirical research is needed to better delineate new professional profiles within real-world service contexts.

Service Design Process

The service design process is also changing to accommodate the unique characteristics of human-AI interactions. Each quadrant of the Hybrid Service Encounter introduces specific shifts in how services are conceived, prototyped, and refined, thus requiring designers to revisit established practices.

Human-to-Human. In this quadrant, the hidden integration of AI demands new design methods to assess what degree of AI involvement should remain hidden or be made transparent to service users and to prototype services where part of the delivery is silently shaped by algorithms. Designers also face questions about how to represent AI components and their workflows in service blueprints as well as how to evaluate their impact on service quality, empathy, and trust.

Human-to-AI. When AI acts as the service user, the design process shifts dramatically. Here, designers need to create data models and decision-making frameworks tailored to machine logic—for example, specifying the types of datasets AI will use, the rules it should follow, and the limits within which it should operate. This might imply rethinking journey maps in machine-readable formats so that AI systems can autonomously navigate, make decisions, and trigger service actions. This quadrant also challenges designers to reimagine user needs and goals from a non-human perspective and to articulate what constitutes a valuable interaction for an AI customer.

AI-to-Human. In this quadrant, AI systems provide services often operating autonomously or in co-piloting roles with human agents. Here, the design process must account for real-time responsiveness, contextual adaptation, and ongoing personalization of services. Traditional linear design methods are insufficient here; instead, designers must embrace iterative workflows that can evolve post-launch. Prototyping must simulate adaptive behavior, and blueprinting must visualize how AI

agents interpret and respond to user data over time. AI's ability to learn and update its behavior introduces new requirements also for continuous monitoring, turning service design into an ongoing lifecycle rather than a fixed sequence of phases.

AI-to-AI. This quadrant demands an even more fundamental rethinking of the service design process. Designers are tasked with creating infrastructures that enable autonomous systems to communicate, negotiate, and collaborate without human oversight. This requires a shift toward designing protocols and interoperable data structures in collaboration with other professional figures. Additionally, traditional methods like service blueprints need to include the representation of algorithmic handoffs and decision chains, while also addressing transparency, accountability, and ethical traceability. For example, in autonomous supply chains, one AI agent may detect a demand signal while another fulfills an order—yet designers must still ensure that these exchanges align with broader service values and governance frameworks.

Across all quadrants, the increasing presence of AI also reshapes how designers work. AI tools are being incorporated into the design process to support ideation, analysis, and testing. For instance, designers may use clustering algorithms to uncover user segments, employ natural language processing to analyze qualitative feedback, or prompt generative models to explore alternative concepts (Chromik, Lachner, and Butz 2020; Yildirim et al. 2022). These AI applications can augment creative thinking. Yet they also introduce new dependencies and call for reflective practices that ensure human judgment remains central in critical design decisions. Further, while these opportunities are promising, the field still lacks a comprehensive framework that defines how best to adapt service design methods to AI-driven contexts. Addressing this gap—both in terms of tools used by designers and in terms of the processes through which AI-powered services are developed—will be essential for advancing service design as a practice.

Service Design Outputs

The outputs of service design—whether tangible deliverables such as service touchpoints or intangible components like service experiences—are significantly influenced by the integration of AI redefining creation, evaluation, and ownership.

Human-to-Human. In this quadrant, service outputs remain largely centered on human experiences, but Service AI plays a supporting role in enhancing the delivery of those experiences. Outputs in these cases are linked to service offerings, interactions, and experiences for humans, which typically remain co-created between users and human providers. However, a key implication for designers is determining how visible the AI's contribution should be, ensuring that service interfaces align with user expectations (Yildirim et al. 2022).

Human-to-AI. Here, the role of AI as service user requires new types of service deliverables—such as structured data formats and system-to-system communication protocols—that cater to algorithmic interpretation. For these types of interactions, therefore, designers must rethink interaction protocols and decision-making structures so that AI systems can autonomously act on behalf of humans (Huang and Rust 2022).

AI-to-Human. When AI is the provider and the human is the end-user, service outputs take the form of AI-generated content, recommendations, or actions. Generative AI applications are a prime example in this case. While they support people in content creation, studies show that their contributions typically account for only around 10% of the final output, while humans remain responsible for the remaining 90% (Davenport and Mittal 2022; Perrault and Clark 2024). This demonstrates a co-piloted process that stresses the boundaries of co-creation in services and introduces new questions about originality and intellectual property.

AI-to-AI. In this final quadrant, service design outputs are created for interactions between algorithms. These outputs often take the form of autonomous actions initiated by AI systems (Maglio 2014). For instance, in predictive maintenance scenarios, one AI system might detect early signs of equipment failure and automatically trigger another system to schedule a repair. In such cases, designers must ensure that these machine-generated outputs are compatible and can operate across different platforms through the adoption of interoperability standards, communication protocols, and data formats.

In all quadrants, a consistent challenge is managing the balance between human and machine contribution in shaping service design outputs. Designers increasingly operate as curators and editors, prompting and refining outputs rather than creating them independently. This shift calls for new strategies to evaluate AI contributions, assign ownership, and uphold quality standards—particularly as services become more data-driven, collaborative, and automated.

Conclusions and Limitations

This paper has proposed a conceptual framework—the Hybrid Human–AI Service Encounter—that categorizes service interactions based on whether the provider and user are human or AI. Through this framework, we have presented an agenda for future research and examined the implications for service design in terms of designer roles, processes, and outputs. Through this, we have offered a structured lens to enable service designers and researchers to make sense of the diverse ways AI reshapes services, extending the foundations of service design theory and practice.

As outlined throughout the article, Service AI brings significant benefits to the field: it enhances efficiency, supports personalization, and opens opportunities for designers to

focus on strategic and interpretive tasks. Notably, its use in front- and backstage service elements expands what can be designed and delivered. AI generates content, adapts experiences, and offers predictive insights that support both service delivery and design. These capabilities suggest that designers are no longer the sole creators of solutions but are evolving into co-pilots who collaborate with AI systems to craft experiences. However, these advantages come with limitations. AI's generative capacities—while valuable—are bounded by a lack of emotional understanding and contextual sensitivity. This poses constraints for services that rely on empathy, subtlety, and human judgment. AI often fails to grasp the nuances essential for creating meaningful and inclusive service experiences, where rather human creativity, intuition, and oversight remain indispensable. Additionally, AI-generated content raises unresolved concerns around authorship, originality, and ownership. Since algorithmic outputs are inherently derivative—constructed from pre-trained data—they may not qualify as truly original in the traditional sense. This challenges existing protocols around intellectual property and forces designers and organizations to rethink what constitutes proprietary work. As AI becomes a more active partner in the creative process, establishing fair and transparent frameworks for credit and accountability will be essential. At the same time, designers will need to continuously monitor Service AI's performance as this is only as good as the data it is trained on. Biased or incomplete datasets will inevitably lead to skewed outputs and raise concerns about fairness and inclusivity. To oversee this, designers will need to develop mitigation strategies aimed at ensuring that AI's integration supports ethical and responsible service delivery.

This study contributes to service design in three key ways. First, it foregrounds the distribution of agency in service encounters, a topic that has been underexplored in service design compared to related fields such as service innovation. Second, it offers a way to rethink service design methods (e.g., blueprinting, journey mapping, and prototyping) by accounting for AI actors as both service users and providers. Third, it clarifies the implications of AI beyond the interface level, expanding within underlying service systems. In doing so, the study reinforces the relevance of service design in discussions on AI-powered services while pointing to the need for new capabilities including data literacy, ethical reasoning, and the ability to interpret algorithmic outputs.

This article also has its limitations. The proposed framework was developed through theoretical synthesis and literature analysis. As such, it may reflect the biases and constraints of existing research. Future studies should empirically test the framework, particularly in real-world contexts where human and AI actors interact. Investigating the long-term socio-cultural implications of Service AI, the evolution of service design methods, and the changing nature of designers' competencies will also be vital for refining this conceptual framework further.

Ultimately, while AI offers exciting new avenues for service design, this article encourages to embrace it as a collaborator that enhances—rather than replacing—human potential. By adopting a critical yet constructive stance, service design will be able to better navigate this evolving landscape and shape the future of services in a way that is both technologically advanced and deeply human-centered.

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Supplemental Material

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References

- Andrada, Gloria, Robert W. Clowes, and Paul R. Smart (2023), "Varieties of Transparency: Exploring Agency Within AI Systems," *AI & Society*, 38 (4), 1321-31.
- Bagozzi, Richard P., Michael K. Brady, and Ming-Hui Huang (2022), "AI Service and Emotion," *Journal of Service Research*, 25 (4), 499-504.
- Bauer, Johannes M. and Paulien M. Herder (2009), "Designing Socio-Technical Systems," in *Philosophy of Technology and Engineering Sciences*, Anthonie W. M. Meijers, Dov M. Gabbay, John Woods, and Paul Thagard, eds. Amsterdam, The Netherlands: Elsevier, 601-30.
- Bitner, Mary Jo (1992), "Servicescapes: The Impact of Physical Surroundings on Customers and Employees," *Journal of Marketing*, 56 (2), 57-71.
- Bitner, Mary Jo, Amy L. Ostrom, and Felicia N. Morgan (2008), "Service Blueprinting: A Practical Technique for Service Innovation," *California Management Review*, 50 (3), 66-94.
- Björgvinsson, Erling, Pelle Ehn, and Per-Anders Hillgren (2012), "Agonistic Participatory Design: Working With Marginalized Social Movements," *CoDesign*, 8 (2-3), 127-44.
- Bock, Dora E., Jeremy S. Wolter, and O. C. Ferrell (2020), "Artificial Intelligence: Disrupting What We Know About Services," *Journal of Services Marketing*, 34 (3), 317-34.
- Cautela, Cabirio, Marzia Mortati, Claudio Dell'Era, and Luca Gastaldi (2019), "The Impact of Artificial Intelligence on Design Thinking Practice: Insights from the Ecosystem of Startups," *Strategic Design Research Journal*, 12 (1), 114-34.
- Chromik, Michael, Florian Lachner, and Andreas Butz (2020), "ML for UX?—An Inventory and Predictions on the Use of Machine Learning Techniques for UX Research," in *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society. NordiCHI'20*. New York, NY: Association for Computing Machinery.
- Davenport, Thomas H. and Nitin Mittal (2022), "How Generative AI is Changing Creative Work," (accessed November 14, 2022), (available at: <https://hbr.org/2022/11/how-generative-ai-is-changing-creative-work>).
- Feldman, Sara (Salevati) (2017), "Co-Creation: Human and AI Collaboration in Creative Expression," in *Conference of Electronic Visualisation and the Arts*. London, UK: EVA.
- Forlano, Laura and Anijo Mathew (2017), "From Design Fiction to Design Friction: Speculative and Participatory Design of Values-Embedded Urban Technology," in *Urban Informatics*. London, UK: Routledge, 7-24.
- Girardin, Fabien and Neal Lathia (2017), "When User Experience Designers Partner with Data Scientists," in *AAAI Spring Symposium*, (available at: <https://api.semanticscholar.org/CorpusID:53290616>).
- Glushko, Robert J. (2010), "Seven Contexts for Service System Design," in *Handbook of Service Science*, Paul P. Maglio, Cheryl A. Kieliszewski, and James C. Spohrer, eds. Norwood, MA: Books24x7, 219-49.
- Glushko, Robert J. and Karen Joy Nomorosa (2013), "Substituting Information for Interaction: A Framework for Personalization in Service Encounters and Service Systems," *Journal of Service Research*, 16 (1), 21-38.
- Glynn Mangold, W. and Emin Babakus (1991), "Service Quality: The Front-Stage Versus the Back-Stage Perspective," *Journal of Services Marketing*, 5 (4), 59-70.
- Golder, Peter N., Debanjan Mitra, and Christine Moorman (2012), "What Is Quality? An Integrative Framework of Processes and States," *Journal of Marketing*, 76 (4), 1-23.
- Haenlein, Michael and Andreas Kaplan (2019), "A Brief History of Artificial Intelligence: On the Past, Present, and Future of Artificial Intelligence," *California Management Review*, 61 (4), 5-14.
- Hollebeek, Linda D. and Russell Belk (2021), "Consumers' Technology-Facilitated Brand Engagement and Wellbeing: Positivist TAM/PERMA- Versus Consumer Culture Theory Perspectives," *International Journal of Research in Marketing*, 38 (2), 387-401.
- Huang, Ming-Hui and Roland T. Rust (2018), "Artificial Intelligence in Service," *Journal of Service Research*, 21 (2), 155-72.
- Huang, Ming-Hui and Roland T. Rust (2022), "AI as Customer," *Journal of Service Management*, 33 (2), 210-20.
- Jylkäs, Titta, Mikko Äijälä, Tytti Vuorikari, and Vesaal Rajab (2018), "AI Assistants as Non-Human Actors in Service Design," in *21st DMI: Academic Design Management Conference*. London, UK: Next Wave.
- Kimbell, Lucy (2011), "Designing for Service as One Way of Designing Services," *International Journal of Design*, 5 (2), 41-52.
- Koch, Janin (2017), "Design Implications for Designing with a Collaborative AI," in *2017 AAAI Spring Symposium Series*, (available at: <https://cdn.aaai.org/ocs/15382/15382-68259-1-PB.pdf>).
- Kunneman, Youetta, Mauricy Alves da Motta-Filho, and Jasper van der Waa (2022), "Data Science for Service Design: An Introductory Overview of Methods and Opportunities," *The Design Journal*, 25 (2), 186-204.

- Lee, Dokyun, Kartik Hosanagar, and Harikesh S. Nair (2018), "Advertising Content and Consumer Engagement on Social Media: Evidence from Facebook," *Management Science*, 64 (11), 5105-31.
- Li, Ai Qiang, Nicholas Rich, Pauline Found, Maneesh Kumar, and Steve Brown (2020) "Exploring Product-Service Systems in the Digital Era: A Socio-Technical Systems Perspective," *The TQM Journal*, 32 (4), 897-913.
- Li, Minglong, Dexiang Yin, Hailian Qiu, and Billy Bai (2021), "A Systematic Review of AI Technology-Based Service Encounters: Implications for Hospitality and Tourism Operations," *International Journal of Hospitality Management*, 95, 102930.
- Maglio, Paul P (2014), "Editorial Column—Smart Service Systems," *Service Science*, 6 (1), i-ii.
- Main, Angus and Mick Grierson (2020), "Guru, Partner, or Pencil Sharpener? Understanding Designers' Attitudes Towards Intelligent Creativity Support Tools," *arXiv*, arXiv:2007.04848.
- Manzini, Ezio and Francesca Rizzo (2011), "Small Projects/Large Changes: Participatory Design as an Open Participated Process," *CoDesign*, 7 (3-4), 199-215.
- McCallum, J. Richard and Wayne Harrison (1985), "Interdependence in the Service Encounter," *The Service Encounter: Managing Employee/Customer Interaction in Service Businesses*, 18 (4), 35-48.
- McCormack, Jon, Patrick Hutchings, Toby Gifford, Matthew Yee-King, Maria Teresa Llano, and Mark D'Inverno (2020), "Design Considerations for Real-Time Collaboration With Creative Artificial Intelligence," *Organized Sound*, 25 (1), 41-52.
- Meroni, Anna and Daniela Sangiorgi (2016), *Design for Services*. London, UK: Routledge.
- Ostrom, Amy L., Darima Fotheringham, and Mary Jo Bitner (2019), "Customer Acceptance of AI in Service Encounters: Understanding Antecedents and Consequences," in *Handbook of Service Science*, Vol. II, P. P. Maglio, C. A. Kieliszewski, J. C. Spohrer, K. Lyons, L. Patrício, and Y. Sawatani, eds. Cham: Springer International Publishing, 77-103.
- Patrício, Lia, Raymond P. Fisk, João Falcão e Cunha, and Larry Constantine (2011), "Multilevel Service Design: From Customer Value Constellation to Service Experience Blueprinting," *Journal of Service Research*, 14 (2), 180-200.
- Patrício, Lia, Anders Gustafsson, and Raymond Fisk (2018), "Upframing Service Design and Innovation for Research Impact," *Journal of Service Research*, 21 (1), 3-16.
- Perrault, Raymond and Clark, Jack (2024), *Artificial Intelligence Index Report 2024*. Human-Centered AI Institute, Stanford University: Stanford, CA.
- Pihkala, Suvi and Helena Karasti (2016), "Reflexive Engagement: Enacting Reflexivity in Design and for 'Participation in Plural'," in *Proceedings of the 14th Participatory Design Conference: Full Papers-Volume 1*, Claus Bossen, ed. New York, NY: Association for Computing Machinery, 21-30.
- Rafaeli, Anat, Daniel Altman, Dwayne D. Gremler, Ming-Hui Huang, Dhruv Grewal, Bala Iyer, A. Parasuraman, and Ko de Ruyter (2017), "The Future of Frontline Research: Invited Commentaries," *Journal of Service Research*, 20 (1), 91-9.
- Robinson, Stacey, Chiara Orsingher, Linda Alkire, Arne De Keyser, Michael Giebelhausen, K. Nadia Papamichail, Poja Shams, and Mohamed Sobhy Temerak (2020), "Frontline Encounters of the AI Kind: An Evolved Service Encounter Framework," *Journal of Business Research*, 116 (August), 366-76.
- Rodrigues, Vanessa, Stefan Holmlid, and Johan Blomkvist (2021), "A Designedly Approach to Exploring Disruptions in Service: Insights from Employing a Systems Perspective," *International Journal of Design*, 15 (3), 61-72.
- Roth, Aleda V. and Larry J. Menor (2003), "Insights into Service Operations Management: A Research Agenda," *Production and Operations Management*, 12 (2), 145-64.
- Sangiorgi, Daniela (2011), "Transformative Services and Transformation Design," *International Journal of Design*, 5 (2), 29-40.
- Seidel, Stefan, Nicholas Berente, Aron Lindberg, Kalle Lyytinen, and Jeffrey V. Nickerson (2018), "Autonomous Tools and Design: A Triple-Loop Approach to Human-Machine Learning," *Communications of the ACM*, 62 (1), 50-7.
- Shostack, G. Lynn (1977), "Breaking Free from Product Marketing," *Journal of Marketing*, 41 (2), 73-80.
- Shostack, G. Lynn (1982), "How to Design a Service," *European Journal of Marketing*, 16 (1), 49-63.
- Solomon, Michael R, Carol Surprenant, John A. Czepiel, and Evelyn G. Gutman (1985), "A Role Theory Perspective on Dyadic Interactions: The Service Encounter," *The Journal of Marketing*, 49 (1), 99-111.
- Syam, Niladri and Arun Sharma (2018), "Waiting for a Sales Renaissance in the Fourth Industrial Revolution: Machine Learning and Artificial Intelligence in Sales Research and Practice," *Industrial Marketing Management*, 69 (February), 135-46.
- van der Bijl-Brouwer, Mieke (2022), "Service Designing for Human Relationships to Positively Enable Social Systemic Change," *International Journal of Design*, 16 (1), 23-34.
- van de Poel, Ibo (2020), "Embedding Values in Artificial Intelligence (AI) Systems," *Minds and Machines*, 30 (3), 385-409.
- Van Doorn, Jenny, Martin Mende, Stephanie M. Noble, John Hulland, Amy L. Ostrom, Dhruv Grewal, and J. Andrew Petersen (2017), "Domo Arigato Mr. Roboto: Emergence of Automated Social Presence in Organizational Frontlines and Customers' Service Experiences," *Journal of Service Research*, 20 (1), 43-58.
- Vink, Josina and Kaisa Koskela-Huotari (2022), "Building Reflexivity Using Service Design Methods," *Journal of Service Research*, 25 (3), 371-89.
- Vink, Josina, Kaisa Koskela-Huotari, Bård Tronvoll, Bo Edvardsson, and Katarina Wetter-Edman (2021), "Service Ecosystem Design: Propositions, Process Model, and Future Research Agenda," *Journal of Service Research*, 24 (2), 168-86.
- Wetter-Edman, Katarina, Daniela Sangiorgi, Bo Edvardsson, Stefan Holmlid, Christian Grönroos, and Tuuli Mattelmäki (2014), "Design for Value Co-Creation: Exploring Synergies Between Design for Service and Service Logic," *Service Science*, 6 (2), 106-21.
- Wilson, H. James and Paul R. Daugherty (2018), "Collaborative Intelligence: Humans and AI Are Joining Forces," *Harvard Business Review*, 96 (4), 114-23.
- Xu, Yingzi, Chih-Hui Shieh, Patrick van Esch, and I-Ling Ling (2020), "AI Customer Service: Task Complexity, Problem-Solving

- Ability, and Usage Intention,” *Australasian Marketing Journal*, 28 (4), 189-99.
- Yang, Qian, Nikola Banovic, and John Zimmerman (2018), “Mapping Machine Learning Advances from HCI Research to Reveal Starting Places for Design Innovation,” in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. New York, NY: Association for Computing Machinery, 1-11.
- Yildirim, Nur, Alex Kass, Teresa Tung, Connor Upton, Donnacha Costello, Robert Giusti, et al. (2022), “How Experienced Designers of Enterprise Applications Engage AI as a Design Material,” in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. New York, NY: Association for Computing Machinery.

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